Completeness alone does not give geodesic segments

Conjecture 00000006778

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The official statement assumes only completeness of the space. Under that stated hypothesis it is false. The Hopf–Rinow theorem needs additional geometric hypotheses; these cannot be silently added to a claim about all complete spaces.

Theorem 1. *The complete metric space* $(\mathbb{Z}, d)$*,* $d(m, n) = |m-n|$*, has no constant-speed minimizing geodesic segment from* 0 *to* 1*.*

Proof. A Cauchy sequence of integers is eventually constant: take the Cauchy tolerance 1, and use that any two distinct integers have distance at least 1. Hence $\mathbb{Z}$ is complete.

A minimizing segment parametrized on $[0, 1]$ would have endpoints $\gamma(0) = 0$, $\gamma(1) = 1$ and

$$d(\gamma(s), \gamma(t)) = |s - t|\, d(0, 1).$$

Taking $s = 1/2$ and $t = 0$ gives $|\gamma(1/2)| = 1/2$. The absolute value of an integer is a nonnegative integer, so this is impossible. Reparametrizing an isometric segment on $[0, d(0, 1)]$ gives exactly the same contradiction. □

Scope. This refutes the sentence as written for complete metric spaces. It does not refute Hopf–Rinow for connected finite-dimensional Riemannian manifolds, or the relevant theorems for locally compact length spaces. If such hypotheses were intended, the problem must first state them.

Lean correspondence. The project uses the standard metric and complete-space instance on $\mathbb{Z}$. `IsSegment` gives the precise endpoint and constant-speed minimizing conditions on the real interval subtype. `no_half_distance` proves the integer-distance obstruction, and `no_segment_zero_one` specializes it to the midpoint of a hypothetical segment. There is no assumed absence of paths or imported unproved geometric fact.

Verification. Lean 4.19.0 with Mathlib pinned to `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The accompanying project and verification transcript give the exact theorem and axiom audit. No proof placeholders, native evaluator proof bypass, or custom axiom declarations are used. This is a prepared submission; official acceptance has not been claimed.